

LECTURES ON SPECIAL FUNCTIONS

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1. GAMMA FUNCTION

The function $\Gamma(z)$ is a meromorphic function of z with simple poles at $z = -n$, ($n = 0, 1, 2, \dots$) with respective residue $\frac{(-1)^n}{n!}$. It has a definition by integral representation

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt = \int_0^1 \left[\log \left(\frac{1}{t} \right) \right]^{z-1} dt, \quad \Re(z) > 0.$$

There are some basic function equations

$$\begin{aligned} \Gamma(1+z) &= z\Gamma(z), \quad (\text{Integration by parts}) \\ \Gamma(n+z) &= z(z+1)\dots(z+n-1)\Gamma(z), \end{aligned}$$

so from this we have

$$\lim_{z \rightarrow -m} \frac{\Gamma(n+z)}{\Gamma(z)} = \begin{cases} (-1)^n \frac{m!}{(n-m)!}, & n \leq m, \\ 0, & n > m, \end{cases}$$

which gives the meromorphic continuation and residues.

Some special values of $\Gamma(z)$ are

$$\begin{aligned} \Gamma\left(\frac{1}{2}\right) &= \pi^{\frac{1}{2}}, \quad \Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\pi^{\frac{1}{2}}, \quad \Gamma\left(-\frac{1}{2}\right) = -2\pi^{\frac{1}{2}}, \\ \Gamma\left(\frac{1}{2} + n\right) &= \pi^{\frac{1}{2}} 2^{-2n} \frac{(2n)!}{n!}, \quad \Gamma\left(\frac{1}{2} - n\right) = (-1)^n \pi^{\frac{1}{2}} 2^{2n} \frac{n!}{(2n)!}. \end{aligned}$$

The Pochhammer symbol $(\alpha)_n$ is

$$(\alpha)_n = \alpha(\alpha+1)\dots(\alpha+n-1) = \frac{\Gamma(\alpha+n)}{\Gamma(\alpha)}.$$

The Binomial coefficient $\binom{\alpha}{m}$ is

$$\binom{\alpha}{m} = \frac{\alpha(\alpha-1)\dots(\alpha-m+1)}{m!} = (-1)^m \frac{\Gamma(m-\alpha)}{m!\Gamma(-\alpha)} = \frac{\Gamma(1+\alpha)}{m!\Gamma(1+\alpha-m)}.$$

The multiplication theorem states that

$$\prod_{l=0}^{m-1} \Gamma\left(z + \frac{l}{m}\right) = (2\pi)^{\frac{1}{2}m - \frac{1}{2}} m^{\frac{1}{2} - mz} \Gamma(mz), \quad m = 2, 3, \dots$$

For $m = 2$, it is the duplication formula

$$\Gamma(z)\Gamma\left(z + \frac{1}{2}\right) = 2^{1-2z} \sqrt{\pi} \Gamma(2z).$$

The function $\psi(z)$ is the logarithmic derivative of the gamma function

$$\psi(z) = \frac{d \log \Gamma(z)}{dz} = \frac{\Gamma'(z)}{\Gamma(z)},$$

which is meromorphic with the same simple poles at $z = -n$, ($n = 0, 1, 2, \dots$). The value $-\psi(1) = \gamma$ is the Euler–Mascheroni constant.

2. BETA FUNCTION

The beta function is defined by

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)},$$

By a change of variables, the beta function has the integral representation

$$B(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt = \int_0^1 \frac{t^{x-1} + t^{y-1}}{(1+t)^{x+y}}dt, \quad \Re(x) > 0, \Re(y) > 0$$

The function $B(x, y)$ satisfies the following functional equations

$$\begin{aligned} B(x, y) &= B(y, x), \\ B(x, y+1) &= \frac{y}{x}B(x+1, y) = \frac{y}{x+y}B(x, y), \\ B(x, y)B(x+y, z) &= B(y, z)B(y+z, x). \end{aligned}$$

More generally

$$\begin{aligned} B(x_1, x_2)B(x_1+x_2, x_3)\dots B(x_1+\dots+x_{n-1}, x_n) &= \frac{\Gamma(x_1)\Gamma(x_2)\dots\Gamma(x_n)}{\Gamma(x_1+\dots+x_n)}, \\ [B(n, m)]^{-1} &= m \binom{n+m-1}{n-1} = n \binom{n+m-1}{m-1}, \quad n, m = 1, 2, \dots \end{aligned}$$

Other integral representations include

$$\begin{aligned} B(x, y) &= \int_0^\infty u^{x-1}(1+u)^{-x-y}du \quad (\text{Change of variables}) \\ B(x, y) &= \int_0^1 (v^{x-1} + v^{y-1})(1+v)^{-x-y}dv \end{aligned}$$

Observe that

$$\begin{aligned} B(z, n+1) &= \frac{\Gamma(z)\Gamma(n+1)}{\Gamma(z+n+1)} = \frac{n!}{z(z+1)\dots(z+n)} \\ &= \int_0^1 t^{z-1}(1-t)^n dt = n^{-z} \int_0^n x^{z-1} \left(1 - \frac{x}{n}\right)^n dx, \end{aligned}$$

whereby

$$\lim_{n \rightarrow \infty} \left(1 - \frac{x}{n}\right)^n = e^{-x}.$$

Hence

$$\Gamma(z) = \lim_{n \rightarrow \infty} \frac{n^z n!}{z(z+1)\dots(z+n)}.$$

Thereby

$$\psi(z) = \frac{d \log \Gamma(z)}{dz} = \lim_{n \rightarrow \infty} \left(\log n - \sum_{m=0}^n \frac{1}{z+m} \right).$$

This gives the exact value

$$\gamma = -\psi(1) = \lim_{n \rightarrow \infty} \left(\sum_{m=1}^n \frac{1}{m} - \log n \right) = 0.577215\dots$$

3. TWO APPLICATIONS OF BETA FUNCTION

The function $\psi(z)$ satisfies these function equations

$$\begin{aligned}\psi(z) &= \psi(1+z) - \frac{1}{z}, \\ \psi(1+n) &= 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \gamma \\ \psi(z+n) &= \frac{1}{z} + \frac{1}{z+1} + \cdots + \frac{1}{z+n-1} + \psi(z).\end{aligned}$$

Moreover,

$$\begin{aligned}\psi(mz) &= m^{-1} \sum_{l=0}^{m-1} \psi\left(z + \frac{l}{m}\right) + \log m, \quad m = 1, 2, \dots \\ \psi^{(n)}(mz) &= m^{-n-1} \sum_{l=0}^{m-1} \psi^{(n)}\left(z + \frac{l}{m}\right), \quad n = 1, 2, \dots\end{aligned}$$

We now prove the multiplication theorem. Let

$$F(z) = \prod_{l=0}^{m-1} \Gamma\left(z + \frac{l}{m}\right) / m^{\frac{1}{2}-mz} \Gamma(mz).$$

Therefore

$$\frac{F'(z)}{F(z)} = \sum_{l=0}^{m-1} \psi\left(z + \frac{l}{m}\right) + m \log m - m\psi(mz) = 0.$$

Thus $F(z)$ is independent of z , depending only on m .

Using Stirling formula

$$\Gamma(w) \sim \sqrt{2\pi} w^{w-\frac{1}{2}} e^{-w},$$

and comparing coefficients in both numerator and denominator, we get

$$F(z) = (2\pi)^{\frac{m-1}{2}}.$$

Therefore

$$\prod_{l=0}^{m-1} \Gamma\left(z + \frac{l}{m}\right) = (2\pi)^{\frac{m-1}{2}} m^{\frac{1}{2}-mz} \Gamma(mz).$$

The reflection formula states that

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}.$$

By invoking the beta function

$$\Gamma(z)\Gamma(1-z) = B(z, 1-z) = \int_0^1 t^{z-1}(1-t)^{-z} dt$$

with the substitution

$$t = \frac{u}{1+u}, \quad dt = \frac{du}{(1+u)^2},$$

we obtain

$$\Gamma(z)\Gamma(1-z) = \int_0^\infty \frac{u^{z-1}}{1+u} du = \frac{\pi}{\sin \pi z}. \quad (\text{Residue Theorem})$$

4. RIEMANN ZETA FUNCTION

The function $\zeta(z)$ is a meromorphic function of z and has its only singularity a simple pole, with residue 1, at the point $z = 1$.

$$\begin{aligned}\zeta(z) &= \sum_{n=1}^{\infty} n^{-z}, & \Re(z) > 1 \\ &= [1 - 2^{1-z}]^{-1} \sum_{n=1}^{\infty} (-1)^{n-1} n^{-z}, & \Re(z) > 0 \text{ (Dirichlet test)} \\ &= [1 - 2^{-z}]^{-1} \sum_{n=0}^{\infty} (2n+1)^{-z}, & \Re(z) > 1.\end{aligned}$$

As an infinite product, the function $\zeta(z)$ can be represented as

$$\zeta(z) = \prod_{p \text{ is prime}} (1 - p^{-z})^{-1}.$$

Proof as follows. For each prime p and $\Re(z) > 1$, the geometric series gives

$$(1 - p^{-z})^{-1} = 1 + p^{-z} + p^{-2z} + \dots$$

Hence

$$\prod_{p \text{ is prime}} (1 - p^{-z})^{-1} = \prod_{p \text{ is prime}} (1 + p^{-z} + p^{-2z} + \dots).$$

By the fundamental theorem of arithmetic, every positive integer n has a unique prime factorization

$$n = \prod_{p \text{ is prime}} p^{k_p}$$

where only finitely many k_p are nonzero. Therefore the above expansion contains each term n^{-z} exactly once, and hence

$$\prod_{p \text{ is prime}} (1 - p^{-z})^{-1} = \sum_{n=1}^{\infty} n^{-z} = \zeta(z).$$

A simple proof of the reflection formula is using the infinite product

$$\frac{1}{\Gamma(z)} = ze^{\gamma z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n},$$

Thereby

$$\frac{1}{\Gamma(z)\Gamma(1-z)} = -\frac{1}{z\Gamma(z)\Gamma(-z)} = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

Comparing with the sine product

$$\frac{\sin \pi z}{\pi z} = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right),$$

we have

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}.$$

5. SPECIAL VALUES OF ZETA FUNCTION

The Bernoulli numbers B_n and the Bernoulli polynomials $B_n(x)$ are defined by

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n, \quad \frac{ze^{xz}}{e^z - 1} = \sum_{n=0}^{\infty} \frac{B_n(x)}{n!} z^n.$$

The following expansion is valid for the Bernoulli numbers and polynomials.

$$B_n(x) = \sum_{l=0}^n B_l \binom{n}{l} x^{n-l}, \quad B_n(x+y) = \sum_{m=0}^n \binom{n}{m} B_m(x) y^{n-m}.$$

The Euler numbers E_n and the Euler polynomials $E_n(x)$ are given by

$$\frac{1}{\cosh z} = \frac{2e^z}{e^{2z} + 1} = \sum_{n=0}^{\infty} \frac{E_n}{n!} z^n, \quad |z| < \frac{\pi}{2},$$

$$\frac{2e^{xz}}{e^z + 1} = \sum_{n=0}^{\infty} \frac{E_n(x)}{n!} z^n, \quad |z| < \pi.$$

Thus $E_n = 2^n E_n(\frac{1}{2})$. Relations between Bernoulli and Euler polynomials are

$$B_n(x) = 2^{-n} \sum_{m=0}^n \binom{n}{m} B_{n-m} E_m(2x), \quad n = 0, 1, 2, \dots,$$

$$E_n(x) = \frac{2^{n+1}}{n+1} \left[B_{n+1} \left(\frac{x+1}{2} \right) - B_{n+1} \left(\frac{x}{2} \right) \right]$$

$$= \frac{2}{n+1} \left[B_{n+1}(x) - 2^{n+1} B_{n+1} \left(\frac{x}{2} \right) \right].$$

The Mellin transformation gives

$$\Gamma(z)\zeta(z) = \sum_{n=1}^{\infty} n^{-z} \Gamma(z) = \int_0^{\infty} \frac{t^{z-1}}{e^t - 1} dt$$

$$= \sum_{n=0}^N \frac{B_n}{n!(z+n-1)} + \int_0^1 t^{z-2} \left(\frac{t}{e^t - 1} - \sum_{n=0}^N \frac{B_n}{n!} t^n \right) dt + \int_1^{\infty} \frac{t^{z-1}}{e^t - 1} dt,$$

which leaves the only pole at $z = 1$ with residue $B_0/\Gamma(1) = 1$ since others are cancelled by $1/\Gamma(z)$. Special values of the Riemann zeta function are

$$\zeta(0) = -\frac{1}{2}, \quad \zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \quad \zeta(4) = \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90},$$

$$\zeta(-m) = (-1)^m \frac{B_{m+1}}{m+1}, \quad \zeta(-2m) = 0, \quad m = 1, 2, 3, \dots$$

$$\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m}}{2(2m)!} B_{2m} = \frac{(2\pi)^{2m}}{2(2m)!} |B_{2m}|,$$

$$\zeta(2m+1) = (-1)^{m+1} \frac{(2\pi)^{2m+1}}{2(2m+1)!} \int_0^1 B_{2m+1}(t) \cot(\pi t) dt.$$

From the Poisson summation, we could find

$$(5.1) \quad \zeta(z) = 2^z \pi^{z-1} \sin \frac{\pi z}{2} \Gamma(1-z) \zeta(1-z),$$

where Riemann conjectured that all non-trivial zeros of $\zeta(z)$ lie on $\Re z = \frac{1}{2}$.

6. HYPERGEOMETRIC FUNCTION

Using underline $\underline{a}$ to denote a sequence of parameters $a_1, a_2, \dots$ with given length, the hypergeometric function ${}_pF_q(\underline{a}; \underline{b}; z)$ is a meromorphic function symmetric in its numerator parameters $\underline{a}$ and non-positive integer-valued denominator parameters $\underline{b}$ separately. It has a definition by the series

$${}_pF_q(\underline{a}; \underline{b}; z) = \sum_{k=0}^{\infty} \frac{(a_1)_k \cdots (a_p)_k}{(b_1)_k \cdots (b_q)_k} \frac{z^k}{k!},$$

where common parameters may be cancelled

$${}_{p+1}F_{q+1}(\underline{a}, c; \underline{b}, c; z) = {}_pF_q(\underline{a}; \underline{b}; z).$$

The radius of convergence $|z| < R$ is

$$(1) R = \infty, \quad (p \leq q) \quad (2) R = 1, \quad (p = q + 1) \quad (3) R = 0, \quad (p > q + 1),$$

unless the series terminates.

Limit and integral relations on hypergeometric series are

$$\int_0^{\infty} e^{-t} t^{A-1} {}_pF_q(\underline{a}; \underline{b}; zt) dt = \Gamma(A) {}_{p+1}F_q(A, \underline{a}; \underline{b}; z),$$

$$\int_0^1 t^{A-1} (1-t)^{B-1} {}_pF_q(\underline{a}; \underline{b}; zt) dt = B(A, B) {}_{p+1}F_{q+1}(A, \underline{a}; A+B, \underline{b}; z),$$

$$\lim_{A \rightarrow \infty} {}_{p+1}F_q(A, \underline{a}; \underline{b}; A^{-1}z) = {}_pF_q(\underline{a}; \underline{b}; z),$$

$$\lim_{B \rightarrow \infty} {}_pF_{q+1}(\underline{a}; B, \underline{b}; Bz) = {}_pF_q(\underline{a}; \underline{b}; z).$$

Two particular cases are the Gauss hypergeometric function ${}_2F_1(a, b; c; z)$ and the Kummer confluent hypergeometric function ${}_1F_1(a; c; z)$.

Taking ${}_0F_0(zt) = e^{zt}$ gives Kummer integral

$${}_1F_1(a; b; z) = \frac{1}{B(a, b-a)} \int_0^1 e^{zt} t^{a-1} (1-t)^{b-a-1} dt, \quad \Re(b) > \Re(a) > 0.$$

Taking ${}_1F_0(a; ; zt) = (1-zt)^{-a}$ gives Euler integral for the Gauss function

$${}_2F_1(a, b; c; z) = \frac{1}{B(b, c-b)} \int_0^1 t^{b-1} (1-t)^{c-b-1} (1-zt)^{-a} dt, \quad \Re(c) > \Re(b) > 0..$$

By introducing $\theta = z \frac{d}{dz}$ so that $\theta z^k = k z^k$, these equations are obtained.

$$z(1-z)y'' + [c - (a+b+1)z]y' - aby = 0, \quad (y = {}_2F_1(a, b; c; z)),$$

$$zy'' + (c-z)y' - ay = 0, \quad (y = {}_1F_1(a; c; z)).$$

A power series $f(z) \in \mathbb{Q}[[z]]$ is called globally bounded if, after a rescaling $z \mapsto Nz$, its coefficients have bounded denominators. $f(z)$ is D -finite if it satisfies a nontrivial linear differential equation with polynomial coefficients. Christol conjectured that every globally bounded D -finite series is the diagonal of a multivariate rational function g ,

$$(6.1) \quad f(z) = \text{diag } g(z_1, \dots, z_m) := \sum_{n \geq 0} [z_1^n \cdots z_m^n] g(z_1, \dots, z_m) z^n.$$

A classical unresolved example is ${}_3F_2\left(\frac{1}{9}, \frac{4}{9}, \frac{5}{9}; \frac{1}{3}; z\right)$, which is globally bounded but is not known to be the diagonal of a rational function.

7. CLASSICAL ORTHOGONAL POLYNOMIAL

If some numerator parameter is a non-positive integer $-n, n \in \mathbb{N}$, then $(-n)_k = 0$ for all $k > n$ and hence the hypergeometric series terminates. The classical orthogonal polynomials arise naturally as terminating hypergeometric series.

The Jacobi polynomial is the terminating Gauss hypergeometric function

$$P_n^{(\alpha, \beta)}(x) = \frac{(\alpha + 1)_n}{n!} {}_2F_1 \left(\begin{matrix} -n, & n + \alpha + \beta + 1 \\ & \alpha + 1 \end{matrix}; \frac{1-x}{2} \right).$$

For $\alpha, \beta > -1$, the Jacobi polynomials are orthogonal on $(-1, 1)$ with respect to $w_{\alpha, \beta}(x) = (1-x)^\alpha(1+x)^\beta$. For $\alpha = \beta = \lambda - \frac{1}{2}$, they give the Gegenbauer family

$$C_n^\lambda(x) = \frac{(2\lambda)_n}{(\lambda + \frac{1}{2})_n} P_n^{(\lambda - \frac{1}{2}, \lambda - \frac{1}{2})}(x) = \frac{(2\lambda)_n}{n!} {}_2F_1 \left(\begin{matrix} -n, & n + 2\lambda \\ & \lambda + \frac{1}{2} \end{matrix}; \frac{1-x}{2} \right).$$

Its orthogonality weight is $(1-x^2)^{\lambda - \frac{1}{2}}$. For example,

$$\begin{aligned} (\text{Legendre}) \quad P_n(x) &= P_n^{(0,0)}(x), & w(x) &\equiv 1, \\ (1^{\text{st}} \text{ Chebyshev}) \quad T_n(x) &= \frac{n!}{(\frac{1}{2})_n} P_n^{(-\frac{1}{2}, -\frac{1}{2})}(x), & w(x) &= (1-x^2)^{-1/2}, \\ (2^{\text{nd}} \text{ Chebyshev}) \quad U_n(x) &= \frac{(n+1)n!}{(\frac{3}{2})_n} P_n^{(\frac{1}{2}, \frac{1}{2})}(x), & w(x) &= (1-x^2)^{1/2}. \end{aligned}$$

The Laguerre polynomial is the terminating Kummer series

$$L_n^{(\alpha)}(x) = \frac{(\alpha + 1)_n}{n!} {}_1F_1(-n; \alpha + 1; x).$$

For $\alpha > -1$, the Laguerre polynomials are orthogonal on $(0, \infty)$ with respect to $w_\alpha(x) = x^\alpha e^{-x}$. The Hermite polynomial is also obtained from two terminating Kummer series, according to parity. For even degree,

$$H_{2n}(x) = (-1)^n \frac{(2n)!}{n!} {}_1F_1 \left(-n; \frac{1}{2}; x^2 \right),$$

and for odd degree,

$$H_{2n+1}(x) = (-1)^n \frac{2(2n+1)!}{n!} x {}_1F_1 \left(-n; \frac{3}{2}; x^2 \right).$$

If the weight w satisfies the Pearson equation

$$(\sigma w)' = \tau w, \quad \deg \sigma \leq 2, \quad \deg \tau \leq 1.$$

then the differential operator $T(y)(x) = \sigma(x)y'' + \tau(x)y' + \lambda_n y$ is formally self-adjoint under suitable boundary conditions. The Rodrigues formula is

$$(7.1) \quad P_n(x) = \frac{C_n}{w(x)} \frac{d^n}{dx^n} [\sigma(x)^n w(x)],$$

which determines orthogonality. The Bochner–Krall problem seeks to classify all orthogonal polynomial systems in terms of high-order differential operators, which still remains open.

TABLE 1. Classical orthogonal polynomial families and their data.

Family	Type	Interval	Weight $w(x)$	$\sigma(x)$	$\tau(x)$
Jacobi	${}_2F_1$	$(-1, 1)$	$(1-x)^\alpha(1+x)^\beta$	$1-x^2$	$\beta - \alpha - (\alpha + \beta + 2)x$
Gegenbauer	${}_2F_1$	$(-1, 1)$	$(1-x^2)^{\lambda-\frac{1}{2}}$	$1-x^2$	$-(2\lambda+1)x$
Legendre	${}_2F_1$	$(-1, 1)$	1	$1-x^2$	$-2x$
1 st Chebyshev	${}_2F_1$	$(-1, 1)$	$(1-x^2)^{-\frac{1}{2}}$	$1-x^2$	$-x$
2 nd Chebyshev	${}_2F_1$	$(-1, 1)$	$(1-x^2)^{\frac{1}{2}}$	$1-x^2$	$-3x$
Laguerre	${}_1F_1$	$(0, \infty)$	$x^\alpha e^{-x}$	x	$\alpha + 1 - x$
Hermite	${}_1F_1$	$\mathbb{R}$	e^{-x^2}	1	$-2x$

8. JACK POLYNOMIAL

Let $\mathbb{Q}(\alpha)$ be the field of fractions of the polynomial ring $\mathbb{Q}[\alpha]$ with rational coefficients in $\mathbb{Q}$ and indeterminate α . If a partition λ has m_i parts equal to i , then write $z_\lambda = (1^{m_1} 2^{m_2} \dots) m_1! m_2! \dots$. Define a bilinear scalar product on the vector space of symmetric functions $\Lambda(\alpha) = \Lambda \otimes \mathbb{Q}(\alpha)$ with indeterminates in $x = (x_1, x_2, \dots)$ (possibly infinite) and coefficients in $\mathbb{Q}(\alpha)$ by this condition

$$\langle p_\lambda, p_\mu \rangle_\alpha = \delta_{\lambda\mu} z_\lambda \alpha^{\ell(\lambda)},$$

where p_λ is the power-sum symmetric polynomial and $\ell(\lambda)$ is the length of λ .

The system of J-normalised Jack symmetric polynomials $J_\lambda = J_\lambda(x; \alpha)$ satisfies the following axioms

- (J1) (Orthogonality) $\langle J_\lambda, J_\mu \rangle_\alpha = 0$ if $\lambda \neq \mu$;
- (J2) (Triangularity) Expanding them in terms of monomials

$$J_\lambda(x; \alpha) = \sum_{\mu} v_{\lambda\mu}(\alpha) m_\mu(x),$$

$v_{\lambda\mu}(\alpha) = 0$ unless $\mu \leq \lambda$ in the dominance ordering.

- (J3) (Normalization) If $|\lambda| = n$, then the coefficient $v_{\lambda, 1^n}$ in $J_\lambda(x; \alpha)$ is $n!$.

If $(i, j) \in \lambda$, then the quantity $h_\lambda(i, j) = h(i, j) = \lambda_i + \lambda'_j - i - j + 1$ is called the hook length at (i, j) . We define two α -refinements of $h_\lambda(i, j)$ as follows

$$\begin{aligned} h_\lambda^*(i, j) &= h^*(i, j) = \lambda'_j - i + \alpha(\lambda_i - j + 1) \\ h_\lambda^\lambda(i, j) &= h_*(i, j) = \lambda'_j - i + 1 + \alpha(\lambda_i - j). \end{aligned}$$

The P-normalised Jack polynomial $P_\lambda = P_\lambda(x; \alpha)$ is related to $J_\lambda(x; \alpha)$ by

$$J_\lambda(x; \alpha) = \left(\prod_{s \in \lambda} h_\lambda^\lambda(s) \right) P_\lambda(x; \alpha)$$

Furthermore, the norm square of J_λ is given by the formula $j_\lambda(\alpha) = \langle J_\lambda, J_\lambda \rangle_\alpha = \prod_{s \in \lambda} h_\lambda^\lambda(s) h_\lambda^*(s)$, so the C -normalized Jack polynomial $C_\lambda = C_\lambda(x; \alpha)$ is related to $J_\lambda(x; \alpha)$ by

$$C_\lambda(x; \alpha) = \frac{\alpha^{|\lambda|} |\lambda|!}{j_\lambda(\alpha)} J_\lambda(x; \alpha).$$

When $\alpha = 1$, $s_\lambda(x) = P_\lambda(x; 1)$ reduces to the Schur polynomial and when $\alpha = 2$, $C_\lambda(x) = C_\lambda(x; 2)$ is the classical zonal polynomial.

9. MACDONALD POLYNOMIAL

The Hall–Littlewood polynomials are defined by the same orthogonality and triangularity principles as the Jack polynomials, but with the scalar product

$$\langle p_\lambda, p_\mu \rangle_t = \delta_{\lambda\mu} z_\lambda(t), \quad z_\lambda(t) = z_\lambda \prod_{i=1}^{\ell(\lambda)} \frac{1}{1 - t^{\lambda_i}}.$$

Thus the factor associated with each part r is $(1 - t^r)^{-1}$, whereas for the Jack product it is the constant α :

$$\langle p_\lambda, p_\mu \rangle_\alpha = \delta_{\lambda\mu} z_\lambda \alpha^{\ell(\lambda)}.$$

Macdonald introduced the two-parameter scalar product

$$\langle p_\lambda, p_\mu \rangle_{q,t} = \delta_{\lambda\mu} z_\lambda(q, t), \quad z_\lambda(q, t) = z_\lambda \prod_{i=1}^{\ell(\lambda)} \frac{1 - q^{\lambda_i}}{1 - t^{\lambda_i}},$$

which replaces the Hall–Littlewood factor $(1 - t^r)^{-1}$ by

$$\frac{1 - q^r}{1 - t^r}.$$

The system of P -normalized Macdonald symmetric polynomials $P_\lambda = P_\lambda(x; q, t)$ is characterized by the following axioms:

- (M1) (Orthogonality) $\langle P_\lambda, P_\mu \rangle_{q,t} = 0$, for $\lambda \neq \mu$.
- (M2) (Triangularity) Expanding them in terms of monomials,

$$P_\lambda(x; q, t) = \sum_{\mu} u_{\lambda\mu}(q, t) m_\mu(x),$$

one has $u_{\lambda\mu}(q, t) = 0$ unless $\mu \leq \lambda$ in the dominance ordering.

- (M3) (P -normalization) The coefficient of m_λ in $P_\lambda(x; q, t)$ is $u_{\lambda\lambda}(q, t) = 1$.

For $q = 0$, the Macdonald product reduces to the Hall–Littlewood product,

$$\langle p_\lambda, p_\mu \rangle_{0,t} = \delta_{\lambda\mu} z_\lambda \prod_{i=1}^{\ell(\lambda)} \frac{1}{1 - t^{\lambda_i}},$$

and hence

$$P_\lambda(x; 0, t) = P_\lambda(x; t).$$

To obtain the Jack polynomials, put $q = t^\alpha$ and let $t \rightarrow 1$. Since

$$\frac{1 - t^{\alpha r}}{1 - t^r} \rightarrow \alpha, \quad r \geq 1,$$

we have

$$\langle p_\lambda, p_\mu \rangle_{t^\alpha, t} \rightarrow \delta_{\lambda\mu} z_\lambda \alpha^{\ell(\lambda)},$$

and therefore

$$P_\lambda(x; \alpha) = \lim_{t \rightarrow 1} P_\lambda(x; t^\alpha, t).$$

Thus the Macdonald scalar product simultaneously extends the Hall–Littlewood product and degenerates to the Jack product.

10. BESSEL FUNCTION

The normalized Bessel function

$$j_\alpha(z) = {}_0F_1\left(\alpha + 1; -\frac{z^2}{4}\right) = 2^\alpha \Gamma(\alpha + 1) z^{-\alpha} J_\alpha(z)$$

satisfies the classical Sonine formula

$$j_{\alpha+\beta}(z) = 2 \frac{\Gamma(\alpha + \beta + 1)}{\Gamma(\alpha + 1)\Gamma(\beta)} \int_0^1 j_\alpha(zx) x^{2\alpha+1} (1-x^2)^{\beta-1} dx, \quad \alpha > -1, \quad \beta > 0.$$

It expresses a Bessel function of higher index as a positive average of Bessel functions of lower index.

More generally, let $R \subset \mathbb{R}^n$ be a reduced root system, W its reflection group, and $k \geq 0$ a W -invariant multiplicity. The Dunkl operators are

$$T_\xi(k)f(x) = \partial_\xi f(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, \xi \rangle \frac{f(x) - f(\sigma_\alpha x)}{\langle \alpha, x \rangle}.$$

Their joint exponential eigenfunction is the Dunkl kernel E_k , and the associated generalized Bessel function is its Weyl symmetrization

$$J_k(x, z) = \frac{1}{|W|} \sum_{w \in W} E_k(wx, z).$$

For $R = \{\pm 1\}$, this reduces to $J_k(x, z) = j_{k-\frac{1}{2}}(ixz)$, and the positive intertwining formula between two multiplicities is precisely the classical Sonine formula.

The first genuinely multivariate case considered here is

$$B_n = \{\pm e_i, \pm e_i \pm e_j : 1 \leq i < j \leq n\}, \quad W(B_n) = S_n \ltimes \mathbb{Z}_2^n.$$

A multiplicity is determined by

$$k = (k_1, k_2), \quad k_1 = k(\pm e_i), \quad k_2 = k(\pm e_i \pm e_j).$$

The corresponding Bessel function has the Jack hypergeometric form

$$J_k^B(z, w) = {}_0F_1^{(\alpha)}\left(a; \frac{z^2}{2}, \frac{w^2}{2}\right) = \sum_{\lambda} \frac{C_\lambda^{(\alpha)}(x) C_\lambda^{(\alpha)}(y)}{[a]_\lambda^{(\alpha)} |\lambda|! C_\lambda^\alpha(\mathbf{1})},$$

where $\alpha = k_2^{-1}$, $a = k_1 + k_2(n-1) + \frac{1}{2}$, and the Pochhammer symbol is defined as

$$[a]_\lambda^{(\alpha)} = \prod_{1 \leq i \leq \ell(\lambda)} \left(a - \frac{i-1}{\alpha}\right)_{\lambda_i}$$

For type B_n , the multivariate Sonine problem asks for a similar formula. Put $k = (k_1, k_2)$, $k' = (k_1 + h, k_2)$, $k_1 \geq 0$, $k_2 > 0$, $h > 0$. A necessary condition is

$$h \in \Sigma(k_2) := \left\{ \{0, k_2, \dots, k_2(n-1)\} \dot{-} \mathbb{Z}_+ \right\} \cup (k_2(n-1), \infty),$$

where $\dot{-}$ is the Minkowski minus. For arbitrary $k_1 \geq 0$, $k_2 > 0$, $h > k_2(n-1)$, Rösler and Voit conjectured that there exists a positive probability kernel $\mu_x^{k', k}$ such that

$$(10.1) \quad J_{(k_1+h, k_2)}^B(x, z) = \int J_{(k_1, k_2)}^B(\xi, z) d\mu_x^{k', k}(\xi).$$

Positive complete Sonine formulas are known in $k_2 = \frac{1}{2}, 1, 2$, $h > k_2(n-1)$ and $k_2 > 0$, $h \in k_2 \mathbb{Z}_+$.

11. LEGENDRE FUNCTION

The Legendre polynomial is the terminating Gauss hypergeometric function

$$P_n(x) = {}_2F_1\left(-n, n+1; 1; \frac{1-x}{2}\right).$$

More generally, the normalized Jacobi polynomial is

$$R_n^{(\alpha, \beta)}(x) = {}_2F_1\left(-n, n+\alpha+\beta+1; \alpha+1; \frac{1-x}{2}\right),$$

so that $P_n = R_n^{(0,0)}$. The classical Jacobi–Sonine formula has the form

$$R_n^{(\alpha+\nu, \beta)}(x) = \int_{-1}^1 R_n^{(\alpha, \beta)}(y) d\mu_x(y), \quad \alpha, \beta > -1, \nu > 0,$$

with μ_x a probability measure.

The higher-rank analogue is associated with the non-reduced root system

$$BC_n = \{\pm e_i, \pm 2e_i : 1 \leq i \leq n\} \cup \{\pm e_i \pm e_j : 1 \leq i < j \leq n\}, \quad W = S_n \ltimes \mathbb{Z}_2^n.$$

Let $k = (k_1, k_2, k_3)$ denote the multiplicities on $\pm e_i, \pm 2e_i, \pm e_i \pm e_j$ and put

$$\rho(k) = \frac{1}{2} \sum_{\alpha \in R_+} k(\alpha) \alpha.$$

The commuting trigonometric Cherednik operators are

$$D_\xi(k)f(x) = \partial_\xi f(x) + \sum_{\alpha \in R_+} \frac{k(\alpha) \langle \alpha, \xi \rangle}{1 - e^{-\langle \alpha, x \rangle}} (f(x) - f(\sigma_\alpha x)) - \langle \rho(k), \xi \rangle f(x),$$

and their normalized joint eigenfunction $G(\lambda, k; x)$ is determined by

$$D_\xi(k)G(\lambda, k; x) = \langle \lambda, \xi \rangle G(\lambda, k; x), \quad G(\lambda, k; 0) = 1,$$

with its Weyl symmetrization

$$F_{BC_n}(\lambda, k; x) = \frac{1}{|W|} \sum_{w \in W} G(\lambda, k; wx)$$

being the Heckman–Opdam hypergeometric function. The normalized Heckman–Opdam Jacobi polynomials are $R_\lambda(k; x) = F_{BC_n}(\lambda + \rho(k), k; ix)$, for dominant integral weights λ . For $n = 1$, this reduces to the classical Jacobi family,

$$R_m(k; t) = R_m^{(\alpha, \beta)}(\cos t), \quad \alpha = k_1 + k_2 - \frac{1}{2}, \quad \beta = k_2 - \frac{1}{2},$$

so the Legendre polynomials occur as the rank-one case $\alpha = \beta = 0$.

Put $k' = (k_1 + h_1, k_2 + h_2, k_3)$, where $k_1, k_2, h_1, h_2 \geq 0$ and $k_3 > 0$. A natural BC_n -Jacobi Sonine conjecture states if $h_1 + h_2 > k_3(n-1)$, then for every x there exists a probability measure $m_x^{k', k}$ such that

$$(11.1) \quad R_\lambda(k'; x) = \int R_\lambda(k; y) dm_x^{k', k}(y)$$

for every dominant integral weight λ . Equivalently, in terms of the Heckman–Opdam hypergeometric function,

$$(11.2) \quad F_{BC_n}(\lambda + \rho(k'), k'; ix) = \int F_{BC_n}(\lambda + \rho(k), k; iy) dm_x^{k', k}(y).$$

12. MULTIVARIATE GAMMA FUNCTION I

Let $\mathbb{S}_p$ denote the space of real symmetric $p \times p$ matrices and $\mathbb{S}_p^+, \mathbb{S}_p^{++}$ its positive semidefinite and positive definite cones. Write $\text{etr}(X) = \exp(\text{tr } X)$. The multivariate gamma function is defined by

$$\Gamma_p(a) = \int_{\mathbb{S}_p^{++}} \text{etr}(-X) |X|^{a - \frac{p+1}{2}} dX, \quad \Re(a) > \frac{p-1}{2}.$$

For $p = 1$ this reduces to the ordinary gamma function. A Cholesky change of variables gives the product formula

$$\Gamma_p(a) = \pi^{\frac{p(p-1)}{4}} \prod_{j=1}^p \Gamma\left(a - \frac{j-1}{2}\right).$$

In particular,

$$\Gamma_p(a+1) = \prod_{j=1}^p \left(a - \frac{j-1}{2}\right) \Gamma_p(a).$$

The associated multivariate beta function is

$$B_p(a, b) = \frac{\Gamma_p(a) \Gamma_p(b)}{\Gamma_p(a+b)}.$$

If $T \in \mathbb{S}_p^+$, the Laplace transform of $\text{etr}(-X) |X|^{a - \frac{p+1}{2}}$ is

$$\int_{\mathbb{S}_p^{++}} \text{etr}(-(I+T)X) |X|^{a - \frac{p+1}{2}} dX = \Gamma_p(a) |I+T|^{-a}.$$

Thus the ordinary identity

$$\int_0^\infty e^{-tx} x^{a-1} dx = \Gamma(a) t^{-a}$$

extends from the positive half-line to the domain where $I+T$ is positive definite.

For a function f defined on an open convex cone C , let $D_u f(x)$ denote its directional derivative in the direction $u \in C$. The function f is called completely monotone on C if $f \in C^\infty(C)$ and

$$(-1)^k D_{u_1} \cdots D_{u_k} f(x) \geq 0$$

for every $k \geq 0$, $x \in C$, and $u_1, \dots, u_k \in C$. In one dimension this is the usual condition $(-1)^k f^{(k)}(x) \geq 0$.

The Bernstein–Hausdorff–Widder theorem states that complete monotonicity is equivalent to the existence of a positive measure on the dual cone

$$f(T) = \int_{C^*} e^{-\langle T, X \rangle} \mu(dX).$$

For the cone $\mathbb{S}_p^{++}$, the closure $\overline{\mathbb{S}_p^{++}} = \mathbb{S}_p^+$ and the trace pairing $\langle T, X \rangle = \text{tr}(TX)$ identifies the cone with its dual. Consequently,

$$f(T) = \int_{\mathbb{S}_p^+} \text{etr}(-TX) \mu(dX)$$

is a Laplace transform if and only if f is completely monotone.

13. MULTIVARIATE GAMMA FUNCTION II

For $\Sigma \in \mathbb{S}_p^{++}$, define

$$\mathcal{L}_{a,\Sigma}(T) = |I + \Sigma T|^{-a}, \quad T \in \mathbb{S}_p^+.$$

If T is restricted to diagonal matrices

$$T = \text{diag}(t_1, \dots, t_p), \quad t_i \geq 0,$$

introduce the admissible set

$$\mathcal{A}(\Sigma) = \{a > 0 : |I + \Sigma T|^{-a} \text{ is completely monotone on } \mathbb{R}_+^p\}.$$

The multivariate gamma conjecture states that for every $\Sigma \in \mathbb{S}_p^{++}$ there exists $a_*(\Sigma) \in [0, (p-1)/2]$ such that

$$(13.1) \quad \mathcal{A}(\Sigma) = \frac{1}{2}\mathbb{N} \cup [a_*(\Sigma), \infty).$$

A nonsingular matrix M is called an M -matrix if its off-diagonal entries are non-positive and M^{-1} has non-negative entries. the Bapat criterion states that

$$|I + \Sigma T|^{-1/2}$$

is infinitely divisible if and only if there is a signature matrix $D = \text{diag}(\pm 1, \dots, \pm 1)$ such that $D\Sigma^{-1}D$ is an M -matrix. In this case $\mathcal{L}_{a,\Sigma}(T)$ is a Laplace transform for every $a > 0$. Moreover, $a_*(\Sigma) = 0$ exactly in the infinitely divisible case characterized by the Bapat M -matrix condition.

The same question appears naturally for matrix quadratic forms. Let $A \in \mathbb{S}_n^+$ and $B \in \mathbb{S}_p^+$. Since

$$\text{tr}(BX^TAX) = \text{vec}(X)^T(B \otimes A)\text{vec}(X),$$

if $a_1, \dots, a_n$ are the eigenvalues of A , define

$$\mathcal{L}_{\alpha,A}(B) = |I_{np} + B \otimes A|^{-\alpha} = \prod_{i=1}^n |I_p + a_i B|^{-\alpha}.$$

If $A = cP$ with P an orthogonal projection of rank r , then

$$\mathcal{L}_{\alpha,A}(B) = |I_p + cB|^{-r\alpha}.$$

The Gindikin–Wallach set

$$\Theta_p = \left\{0, \frac{1}{2}, \dots, \frac{p-2}{2}\right\} \cup \left[\frac{p-1}{2}, \infty\right)$$

therefore gives the exact criterion

$$\mathcal{L}_{\alpha,A} \text{ is a Laplace transform} \iff r\alpha \in \Theta_p.$$

This suggests the following refinement.

Let $A \geq 0$ have distinct positive eigenvalues $a_1, \dots, a_q$ with multiplicities $m_1, \dots, m_q$. Another conjecture is the map

$$(13.2) \quad B \mapsto \prod_{i=1}^q |I_p + a_i B|^{-m_i \alpha}$$

is completely monotone on $\mathbb{S}_p^{++}$ if and only if $m_i \alpha \in \Theta_p$ for every $i = 1, \dots, q$.

14. ZONAL POLYNOMIAL I

Let $G = GL_n(\mathbb{R})$ and $K = O(n)$. The group G acts transitively on the cone $\mathbb{S}_n^+$ of positive definite symmetric matrices by congruence

$$G \curvearrowright \mathbb{S}_n^+, \quad g \cdot X = gXg^T,$$

and the stabilizer of I_n is K . Hence

$$\mathbb{S}_n^+ \cong G/K, \quad gK \mapsto gg^T.$$

If $g = k_1 a k_2$ is a singular value decomposition, with $a = \text{diag}(a_1, \dots, a_n)$, then every bi- K -invariant function on G depends only on $a_1^2, \dots, a_n^2$, equivalently on the eigenvalues of $g^T g$.

The occurrence of even partitions can be seen directly from principal minors. After complexification, let $\Delta_r(X)$ denote the determinant of the leading $r \times r$ principal block and, for a partition $\lambda = (\lambda_1, \dots, \lambda_n)$, put

$$\Delta_\lambda(X) = \Delta_1(X)^{\lambda_1 - \lambda_2} \dots \Delta_n(X)^{\lambda_n}.$$

If b is lower triangular with diagonal part $t = \text{diag}(t_1, \dots, t_n)$, then

$$\Delta_\lambda(bXb^T) = t_1^{2\lambda_1} \dots t_n^{2\lambda_n} \Delta_\lambda(X).$$

Thus Δ_λ is a highest-weight polynomial of weight 2λ .

A pair (G, K) is a Gelfand pair if the convolution algebra $L^1(K \backslash G / K)$ is commutative. Equivalently, the quasi-regular representation on $L^2(G/K)$ is multiplicity-free, or

$$\dim V_\pi^K \leq 1$$

for every irreducible representation V_π of G .

The pair $(GL_n(\mathbb{R}), O(n))$ is a Gelfand pair. Its polynomial spherical representations are precisely those of even highest weight,

$$\dim V_\mu^K = \begin{cases} 1, & \mu = 2\lambda, \\ 0, & \text{otherwise.} \end{cases}$$

Choosing a unit K -fixed vector $v_K \in V_{2\lambda}$, the normalized spherical function is

$$\varphi_\lambda(g) = \langle \pi_{2\lambda}(g)v_K, v_K \rangle, \quad \varphi_\lambda(e) = 1,$$

and satisfies

$$\int_K \varphi_\lambda(gkh) dk = \varphi_\lambda(g)\varphi_\lambda(h).$$

Since φ_λ is bi- K -invariant, it is a symmetric function of the eigenvalues of $g^T g$. With the usual normalization,

$$\varphi_\lambda(g) = \frac{Z_\lambda(g^T g)}{Z_\lambda(I_n)}.$$

Thus the zonal polynomials are the radial spherical polynomials of $GL_n(\mathbb{R})/O(n)$.

15. ZONAL POLYNOMIAL II

Let $S = S_{2m}$, $\tau = (12)(34) \cdots (2m-1, 2m)$, and its centralizer be

$$H = C_S(\tau) \cong S_2 \wr S_m.$$

Equivalently, H is the stabilizer of the standard perfect matching, so S/H may be identified with the set of perfect matchings of $\{1, \dots, 2m\}$. The pair (S, H) is a Gelfand pair and the Schur–Weyl duality gives, in the stable range,

$$(V^{\otimes 2m})^{O(n)} \cong \bigoplus_{\lambda \vdash m} S^{2\lambda} \cong \text{Ind}_H^S \mathbf{1}.$$

by the $O(n)$ -invariant tensor $u = \sum_{i=1}^n e_i \otimes e_i$.

If ω^λ denotes the spherical function corresponding to $S^{2\lambda}$, the spherical characteristic map sends it to the zonal polynomial,

$$\text{ch}_H(\omega^\lambda) = Z_\lambda.$$

Hence zonal polynomials admit both the radial realization from $GL_n(\mathbb{R})/O(n)$ and the matching realization from S_{2m}/H .

The Jack deformation of this picture is encoded by the connection coefficients $c_{\mu, \nu}^\lambda(\alpha)$, defined by

$$\sum_{\theta} \frac{J_\theta(x; \alpha) J_\theta(y; \alpha) J_\theta(z; \alpha)}{j_\theta(\alpha)} t^{|\theta|} = \sum_{m \geq 0} t^m \sum_{\lambda, \mu, \nu \vdash m} \frac{c_{\mu, \nu}^\lambda(\alpha)}{\alpha^{\ell(\lambda)} z_\lambda} p_\mu(x) p_\nu(y) p_\lambda(z).$$

At $\alpha = 1$ these coefficients come from the class algebra of the symmetric group, while at $\alpha = 2$ they come from the double-coset algebra of the Gelfand pair (S_{2n}, H_n) .

Let $A = \{a_1, \dots, a_m\}$ and $B = \{b_1, \dots, b_m\}$. A matching of $A \sqcup B$ is a partition of these $2m$ vertices into m unordered pairs. If δ_1, δ_2 are two matchings, their union is a disjoint union of even cycles of lengths $2\rho_1, 2\rho_2, \dots$. Define their coset type by

$$\Lambda(\delta_1, \delta_2) = (\rho_1, \rho_2, \dots) \vdash m.$$

Fix the standard matching

$$\varepsilon = \{\{a_1, b_1\}, \dots, \{a_m, b_m\}\}$$

and, for each $\lambda \vdash m$, choose a matching δ_λ such that $\Lambda(\varepsilon, \delta_\lambda) = \lambda$. Then

$$\mathcal{G}_{\mu, \nu}^\lambda = \{\delta : \Lambda(\varepsilon, \delta) = \mu, \quad \Lambda(\delta, \delta_\lambda) = \nu\}$$

is the set of matchings of type (λ, μ, ν) . A matching is called bipartite if every pair has the form $\{a_i, b_j\}$.

The Matching–Jack Conjecture states that, for $b = \alpha - 1$,

$$(15.1) \quad c_{\mu, \nu}^\lambda(1 + b) \in \mathbb{N}[b],$$

and more precisely that there should exist a non-negative integer statistic wt on $\mathcal{G}_{\mu, \nu}^\lambda$ such that

$$(15.2) \quad c_{\mu, \nu}^\lambda(1 + b) = \sum_{\delta \in \mathcal{G}_{\mu, \nu}^\lambda} b^{\text{wt}(\delta)},$$

with $\text{wt}(\delta) = 0$ precisely for bipartite matchings.

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